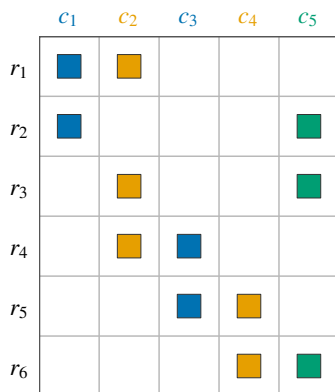
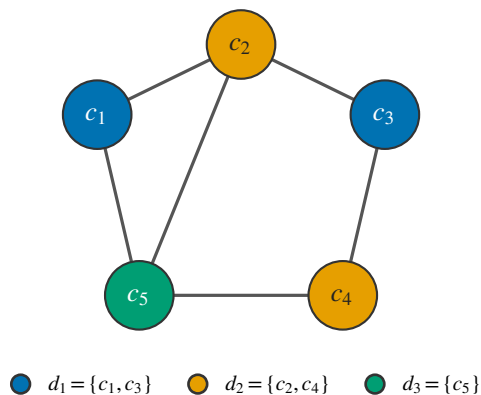


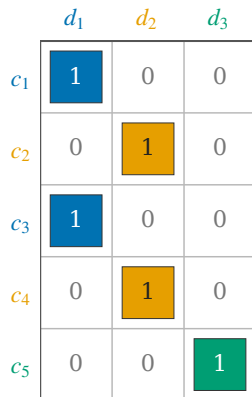
(a) Sparse Jacobian pattern J



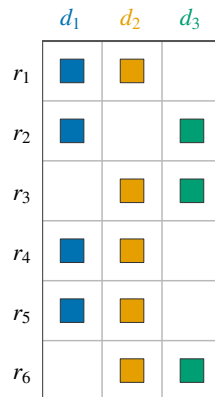
(b) Column-intersection graph G



(c) Seed matrix S



(d) Compressed product JS



Columns assigned to the same direction never share a structural nonzero row.